

WALLSIGHT

RF-Fusion Ambient Sensing — The House That Sees Without Watching

Event: LT HackFest 2026 (International, online) — Round 1 submission Aug 13–15, 2026 **Team:** 3 (System Designer · ML Engineer · Frontend) — Team name: **WallSight** **Hardware budget:** ~₹1,200 · **Software cost:** ₹0 (all open source)

THE ONE-LINER

"Every home already fills itself with Wi-Fi. WallSight turns that radio into a silent guardian — it feels a toddler cross the doorway at 2 AM, hears grandma's breathing through a wall, and alerts your phone. No camera. No mic. No wearable. No phone required."

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1. Executive Summary

WallSight is a device-free human-sensing system built on **WiFi-CSI (Channel State Information)** — the same radio waves every home already has. A set of small plug-and-play boxes (~₹400 each) turns ordinary WiFi signals into a sensor network that can **detect presence, motion, breathing, heart-rate (experimental), room zones, and cross-room movement through walls** — with **no cameras, no microphones, no wearables, and no phone needed from the person being monitored**.

The intelligence lives in the **cloud** (subscription model), so the hardware stays dumb and cheap. Phones and BLE devices in the home become *passive participants* in the sensing ecosystem — the boxes detect them, and their signals vote alongside the WiFi data to confirm what's happening.

Why this wins: every existing safety solution requires the person to participate (wear a watch, carry a phone, stand in front of a camera). WallSight requires **nothing**. That single insight — "device-free sensing, privacy by physics" — is the whole pitch.

2. The Problem

2.1 Who suffers

Person	The failure
Dementia patient	Wanders out at night; wears nothing; family asleep. Watches fail (privacy, no through-wall), wearables get ripped off
Toddler	Crosses toward the kitchen/stairs at 2 AM; no one can watch 24×7
Elderly parent	Falls, or stops breathing in their room; found hours later — the "long lie"
Family	Can't afford 24×7 caregivers; cameras invade everyone's privacy; alerts that need <i>the person to cooperate</i> don't work

2.2 Why existing solutions fail

Solution	Why it fails
Cameras / CCTV	Privacy nightmare (bathroom, bedroom, guests); no through-wall; everyone hates them
Wearables (watches, pendants, tags)	Dementia patients remove them; toddlers won't keep them; batteries die; " judges have seen this a thousand times "
Pressure mats / PIR sensors	Single point, no through-wall, no breathing/heart signal, no tracking, cheap-looking
mmWave radar modules	Slightly better, but still commercial off-the-shelf demos — well-trodden hackathon ground
UWB beacons/tags	₹2,500+/module, needs the person to carry a tag — same participation flaw

2.3 The insight

*Every existing system requires the person to **participate**. WallSight requires **nothing** from the person — the room itself becomes the sensor.*

3. The Physics

3.1 What CSI is

WiFi doesn't travel point-to-point like a laser. It **radiates in all directions** and **bounces off everything** — walls, furniture, and people. The receiver sees not one signal but a *soup of reflected copies* of the signal. **CSI = Channel State Information** — a per-subcarrier measurement of how that soup changes over time (amplitude + phase).

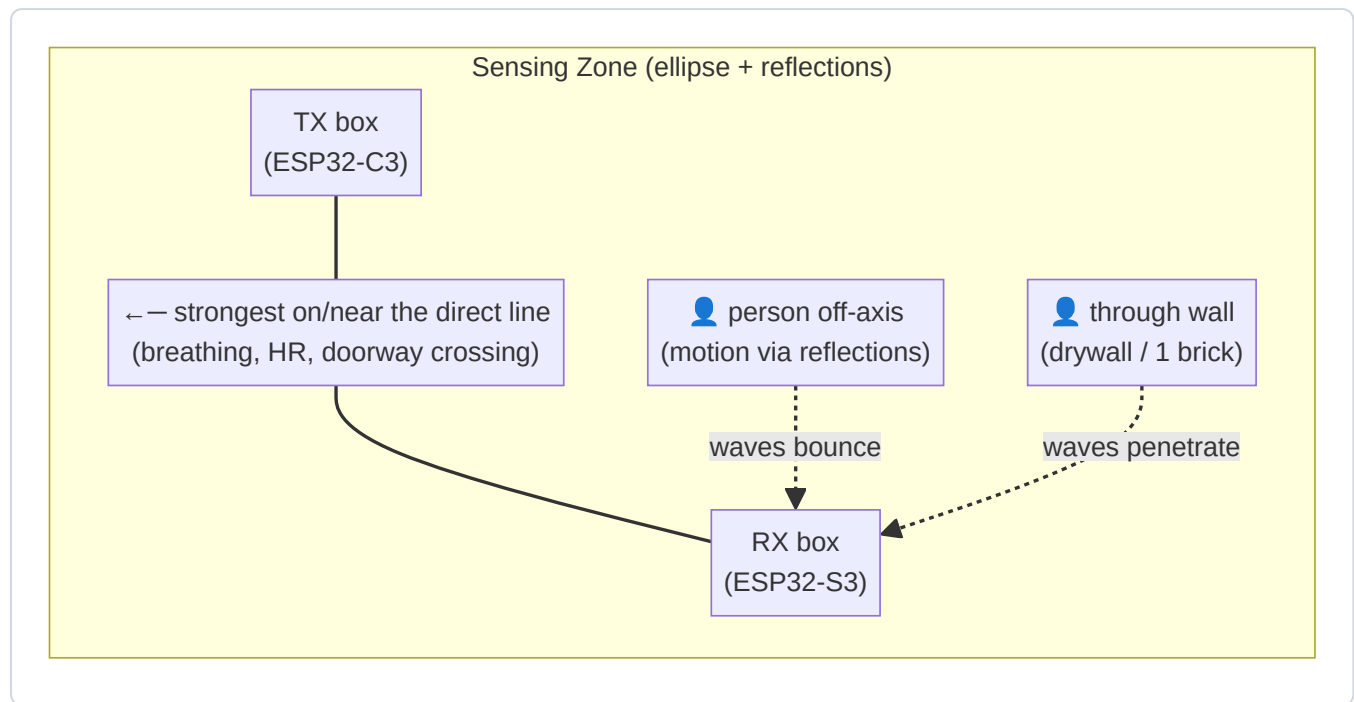
3.2 Why a person is detectable

A moving person **disturbs the reflected paths**:

- **Motion** (walking, waving, breathing chest) → shifts the reflections → measurable CSI changes (Doppler)
- **Static presence near the line** → attenuates the direct path
- **Through walls** → 2.4 GHz penetrates drywall/wood (and 1 brick wall)

An ESP32 receiver capturing CSI at ~10–30 Hz (config-dependent) can see this in real time.

3.3 Sensing geometry — not a line, not a circle, an "egg"

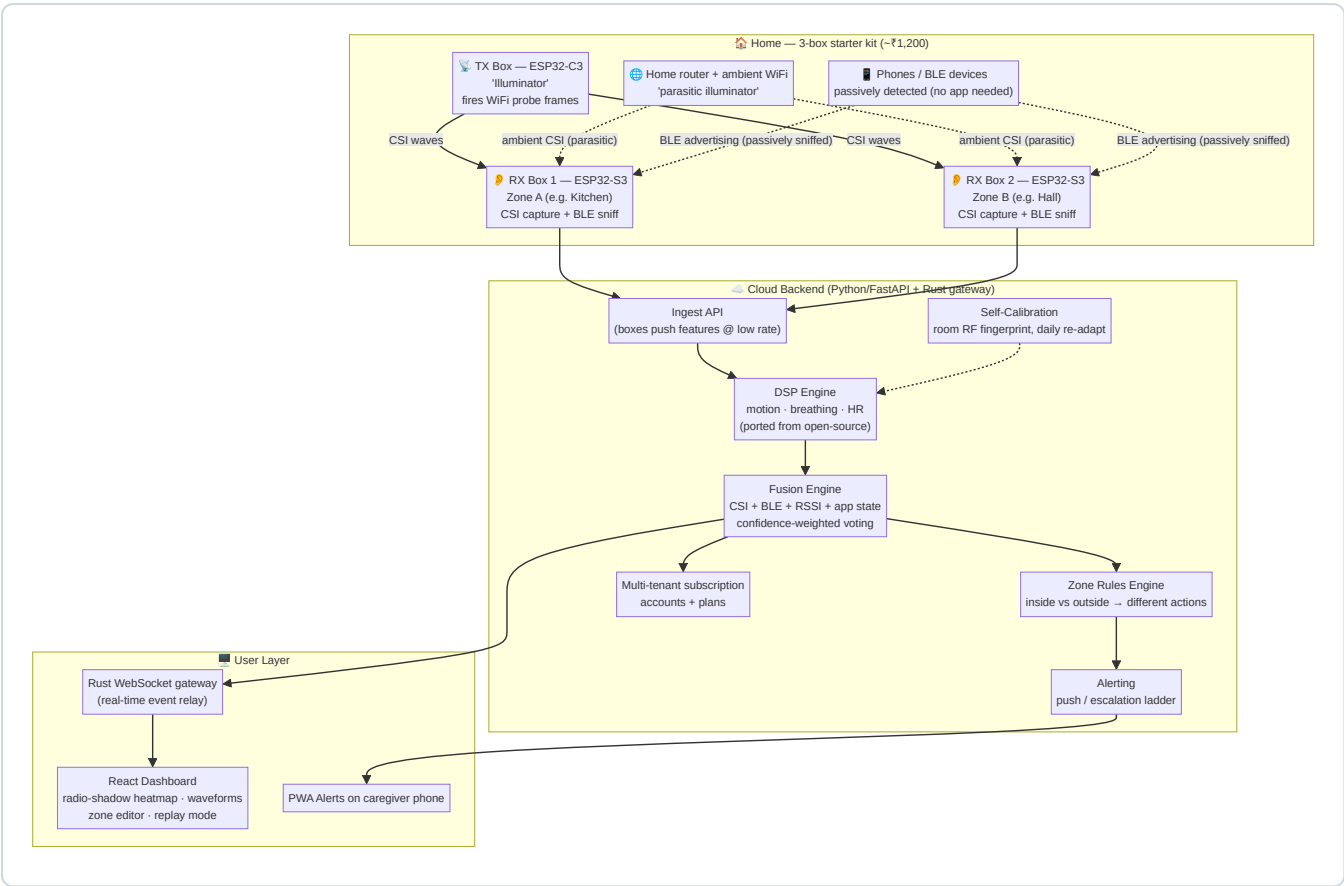


- **On/near the line** → strongest signal (breathing, heart-rate, crossing detection)
- **Anywhere in the room** → detected via reflections (motion)
- **Through walls** → waves penetrate (presence in the next room)
- **Concrete floor slabs** → ❌ block 2.4 GHz — each floor needs its own boxes

3.4 The 30-second explanation (memorize this)

"WiFi signals bounce off everything. When a person moves — even just breathes — they disturb those bounces. WallSight's boxes listen to the radio field and read the disturbance. The person becomes a shadow in the radio waves. We just visualize that shadow."

4. System Architecture



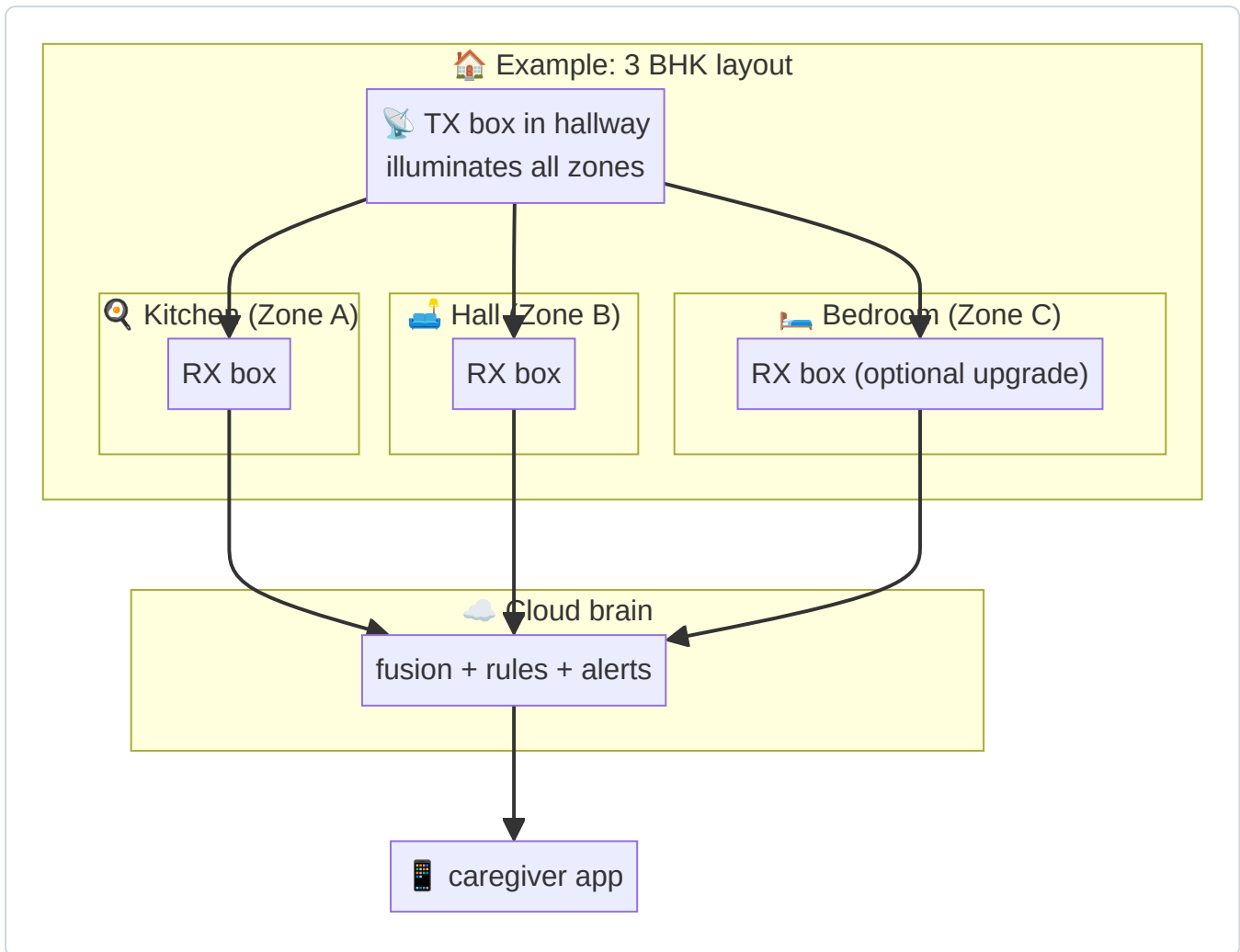
4.1 Key design decisions

Decision	Why
Dumb boxes, smart cloud	₹400 nodes + zero edge compute → subscription business model works; processing scales server-side
3 boxes for Round 1	1 TX + 2 RX = 2 zones → enables zone-tracking + inside/outside behavior demo (minimum for our features)
Python backend (not Rust)	wifi-ghost/ThroughNet algorithms are Python/NumPy — porting to Rust costs 3–4 days we don't have
Rust microservice	A real Rust component (WebSocket gateway) — designer's learning goal, off the critical path
Dashboard works on synthetic data from day 1	Frontend never blocks on hardware; demo never depends on live accuracy
Ready-made firmware, not from scratch	espressif/esp-csi + heyfinal/wifi-ghost ship working ESP32 firmware → "flash + configure + stream" instead of "write"

5. Deployment Model

One box per room/zone, like Wi-Fi extenders. Coverage math:

Home type	Zones	Boxes (RX + TX)	Hardware cost
1 BHK / studio	2	2 RX + 1 TX = 3	~₹1,200
3 BHK Indian (~1,300 sq ft)	5	5 RX + 2 TX = 7	~₹3,000
US home (~2,500 sq ft)	7–9	7–9 RX + 2–3 TX	~₹4,000–5,000
3-storey mansion	per floor × 3	scales linearly (concrete slabs block RF)	~₹1,500/floor



Scaling story (pitch slide): "one box per room, like Wi-Fi extenders — your home's WiFi already lives everywhere; we just make it feel."

6. Innovation Pillars

Pillar 1 — Parasitic Sensing

RX boxes in monitor mode capture CSI from **any** transmitter — the home router, smart TVs, even the neighbor's WiFi bleeding through walls. Our TX box is a *booster*, not a necessity. *We sense using radio that already exists in the walls — including signals we never sent.*

Pillar 2 — Radio-Shadow Visualization ("RF skeleton")

No camera → we render what the radio sees: a **radar-style activity view** of the room — per-zone energy glow, the person as a moving shadow-blob, and a movement trail, with the breathing waveform overlaid. Honest framing: it's a **zone-level energy visualization**, not true spatial localization (2 RX boxes can't triangulate) — we never fake precision. A "skeleton of RF, not pixels."

Pillar 3 — Ecosystem Fusion (every device votes)

CSI boxes (fine) + WiFi RSSI (coarse) + BLE sniff (phones/watches/earbuds) + phone app state (opt-in)

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Cloud fusion engine

each source scores "is someone in zone X" → confidence-weighted vote

conflict resolution: "CSI says hall, BLE says phone in kitchen → engine decides"

Pillar 4 — Self-Calibrating Baseline

Auto-learns each room's RF fingerprint (walls, furniture), re-adapts daily, flags "environment changed" instead of false-alarming. **No install engineer, no calibration app.**

7. Features

Round 1 (must demo — Aug 13–15)

#	Feature	Notes
1	Presence detection, device-free, through walls	Core
2	Motion detection (anywhere in room)	Doppler-based
3	Breathing rate (14–18 BPM)	Open-source verified — star of the demo
4	Doorway crossing detection	Strongest line
5	2-zone occupancy ("Kitchen occupied")	
6	Zone-to-zone tracking (kitchen → hall trail)	
7	BLE device detection ("phone in kitchen")	Passive sniff, no app on that device
8	Inside vs outside zone → different alert actions	Rules engine
9	Radio-shadow activity view	Zone-energy heatmap + movement trail (full render)
10	Self-calibration baseline	
11	Replay mode (recorded CSI)	Demo safety net
12	Local fallback (single box + home router)	
13	Subscription-structured multi-tenant backend	Business model is real code

Round 2 (if we advance)

Feature	Honest status
Heart rate (bandpass 0.8–2.0 Hz FFT)	Experimental — labeled, never faked
Fall-then-stillness heuristic	"sudden motion → zero motion" heuristic, NOT a real fall detector
Apnea alert (breathing stopped)	
Abnormal breathing/HR rate alerts	
Escalation ladder (push → SMS/email → all caregivers)	
Caregiver family groups (multi-user per home)	
Event history timeline (signed audit)	
Night/quiet mode	
Consent-based phone participation + random-MAC handling	
CSV data export	

Roadmap

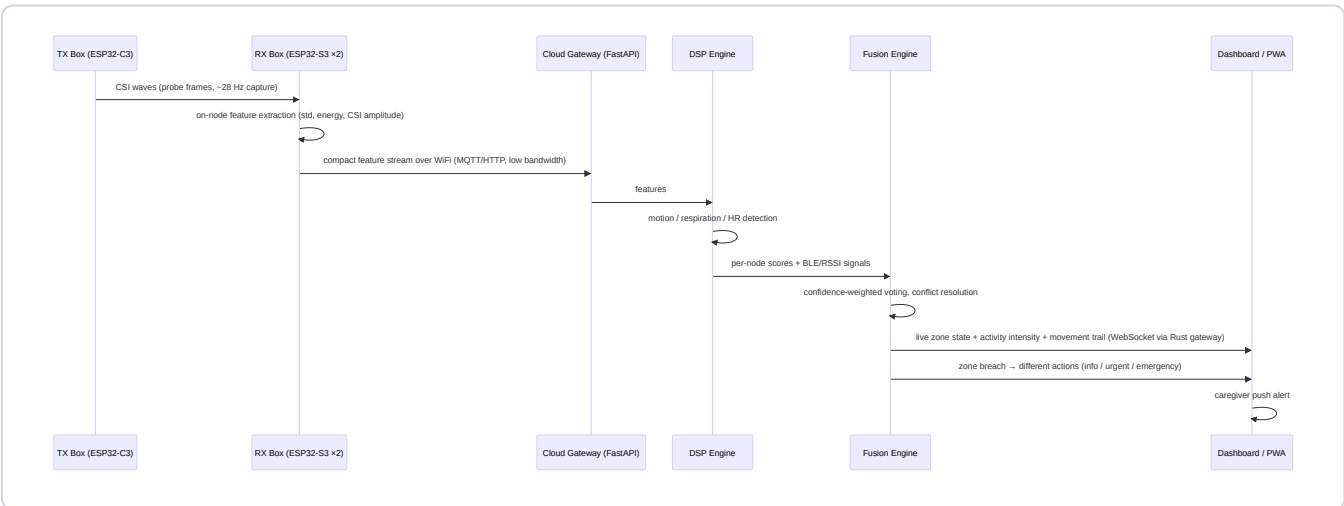
Feature	Notes
OpenWrt router as extra RX node	\$30 router becomes a high-gain sensing node
Co-crystal C3+S3 clock-synced boxes	Espressif reference design — phase data
Room-level triangulation (multi-RX)	
Firmware OTA updates	
Offline buffering	
Multi-home management in one app	

8. Honest Limits

⚠️ *Know these cold. Judges will probe them. Never overclaim — "we know our limits" is a strength.*

Question	Honest answer
"Can it detect a fall?"	Only a <i>sudden-motion-then-stillness</i> heuristic — not a reliable fall detector. Labeled honestly.
"What's the accuracy?"	We publish our own measured numbers (e.g., "94% zone occupancy, 0 false positives in X-hour test").
"Through concrete floors?"	❌ No — 2.4 GHz blocked by slabs. Boxes per floor, like Wi-Fi extenders.
"Through how many walls?"	~1 brick wall reliably; 2 marginal. Drywall (US homes) easier.
"Can it identify WHO?"	❌ No identity from CSI. Optional person-ID only via their phone's BLE MAC (consent-based).
"Is the heatmap a camera?"	❌ It's rendered from radio data — a visualization, not pixels. Nothing is recorded visually.
"Can it localize the person precisely?"	❌ Room-level only . The heatmap is zone-energy + movement trail, not true positioning — 2 RX boxes can't triangulate precisely. We never fake precision.
"RuView already does this — why you?"	✅ We know — 83K stars proves the tech. RuView is a free DIY edge platform (no app, no cloud, no India); we ship the consumer product around it.
"Is heart rate real?"	Research-grade, fragile in live demo → we demo breathing as the star; HR = experimental.
"Range?"	~10–15 ft per TX–RX pair. A per-room system, not outdoor/whole-building.
"Why not a ₹5 PIR sensor?"	PIR: no through-wall, no breathing, no zones, no tracking, no data richness. Different product class.
"Does it need the person to carry anything?"	No. Nothing. That's the entire point.
"Does it work if the person is perfectly still?"	Static presence works near the line ; still-person detection off-axis is weak — honest limit.
"Privacy?"	No camera/mic/wearable. Features-only streaming (raw CSI not stored), encrypted transport, retention controls, consent-based phone participation.

9. Data Pipeline



Bandwidth honesty: boxes transmit compact features, not raw CSI dumps → cheap, battery-friendly, subscription-viable at scale.

10. Hardware & Costs

Item	Qty	Unit cost	Total	Source
ESP32-S3 SuperMini (RX boxes)	2	~₹400	~₹800	Amazon / robu.in / Delhi market
ESP32-C3 SuperMini (TX box)	1	~₹300	~₹300	same
USB-C cables + jumpers	misc	—	~₹100	local
Total out-of-pocket			≈ ₹1,200	

- **Cloud:** ₹0 — free tiers (Render/Railway) + localhost for the demo
- **Software:** ₹0 — 100% open source
- **Total prize pool of the hackathon:** ₹25,45,000 — 1st place ₹2,50,000; top 100 paid (51st–100th = ₹15,000 each)

11. What We Need

Buy (ORDER TODAY — Aug 3 at the latest; 3–5 day shipping)

- 2× ESP32-S3 SuperMini
- 1× ESP32-C3 SuperMini
- USB-C cables

Accounts (free)

- GitHub (public repo for submission)
- Render/Railway (optional — localhost works for demo)

Repos to clone (all open source)

Repo	What we take from it
<code>espressif/esp-csi</code>	Official ESP32 CSI capture (all ESP32 chips supported: ESP32/S2/C3/S3/C5/C6/C61)
<code>heyfinal/wifi-ghost</code>	Working 2× ESP32-WROOM firmware + Python motion/respiration algorithms (verified 14–18 BPM through wall)
<code>Adichapati/ThroughNet</code>	Multi-node CSI fusion reference (100% detection / 0% false positive on ESP32-S3)
<code>ruvnet/RuView</code> (~83K stars, Rust)	Proof the tech works (presence/breathing/HR) — study, don't clone; we build the product around it

Skills in team

Skill	Who
ESP-IDF / C firmware (adapt, not write)	Designer
Python + NumPy/SciPy DSP	ML guy
React / TS dashboard	Frontend
Rust (small gateway service)	Designer
System architecture + integration	Designer
Pitching / video / design	Frontend + Designer

Time

~30 focused hours per person until Aug 13.

12. Business Model

Why subscription

- Hardware is dumb + cheap (₹300–400/node); **all intelligence is cloud-side** → recurring revenue, low COGS, scales to any home size
- The architecture IS the business model — multi-tenant accounts, plans, device groups from day 1

Plans

Plan	Boxes	Price
Starter	3-box	₹199/mo
Family (3 BHK)	7-box	₹499/mo
Estate	N-box	custom

Market

- 40M+ seniors in India living with family; wandering/falls are top caregiver fears
- B2B2C: elder-care providers, assisted-living centers, hospital dementia wards
- Daycares (toddler safety), pet-monitoring adjacent later
- Global: 1.4B+ connected homes; WiFi-sensing commercial players (Origin Wireless) are enterprise-priced, not consumer

COGS

Item	Cost
3 BHK hardware kit	~₹3,000
Server per home	~₹0 (free-tier architecture; near-zero marginal)
Gross margin	~90%+ at ₹499/mo

Competition framing (slide)

"RuView (83K stars) and Origin Wireless proved WiFi sensing works. But RuView is a free DIY platform for hobbyists — English, Home-Assistant world, edge-only, no cloud, no app — and Origin is enterprise-priced. Neither ships a consumer product for families. We're the ₹1,200 + ₹199/mo product: caregiver app, cloud subscription, BLE fusion, 10-minute setup."

13. The Demo

5-minute video script

#	Scene	On-screen
1	Empty room	Dashboard: ALL ZONES CLEAR
2	Person walks in	Shadow-blob appears on heatmap → KITCHEN OCCUPIED
3	Phone placed on table	DEVICE DETECTED: phone in kitchen (BLE)
4	Person sits still	Live breathing waveform: 16 BPM
5	Person crosses doorway	CROSSING EVENT
6	Simulated 2 AM zone breach	Urgent escalation → daughter's phone
7	Close	"No camera. No mic. No wearable. The WiFi in your walls felt everything."

Recording safety

- Every scene rehearsed with **replay mode** (recorded data) → the demo NEVER fails live
- Live shots included but optional in the edit

14. Team Roles

Person	Role	Owns	Deliverable by
System Designer (you)	Architect + integrator	Firmware bring-up (flash/adapt ready-made code), stream format, Rust WebSocket gateway, integration between layers, hardware ordering, demo coordination, pitch narrative, submission	Working 3-box CSI stream by Aug 10
ML guy	Algorithms	Python FastAPI backend: ingest, DSP port (motion/resp/HR), fusion engine, self-calibration, accuracy metrics for pitch	Live backend + measured numbers by Aug 11
Frontend	Product surface	React dashboard (floor-plan editor, heatmap, waveforms, zone editor, alert feed, replay), PWA alerts, demo video editing, deck visuals	Dashboard on synthetic data by Aug 7 , real data by Aug 11 , video by Aug 12

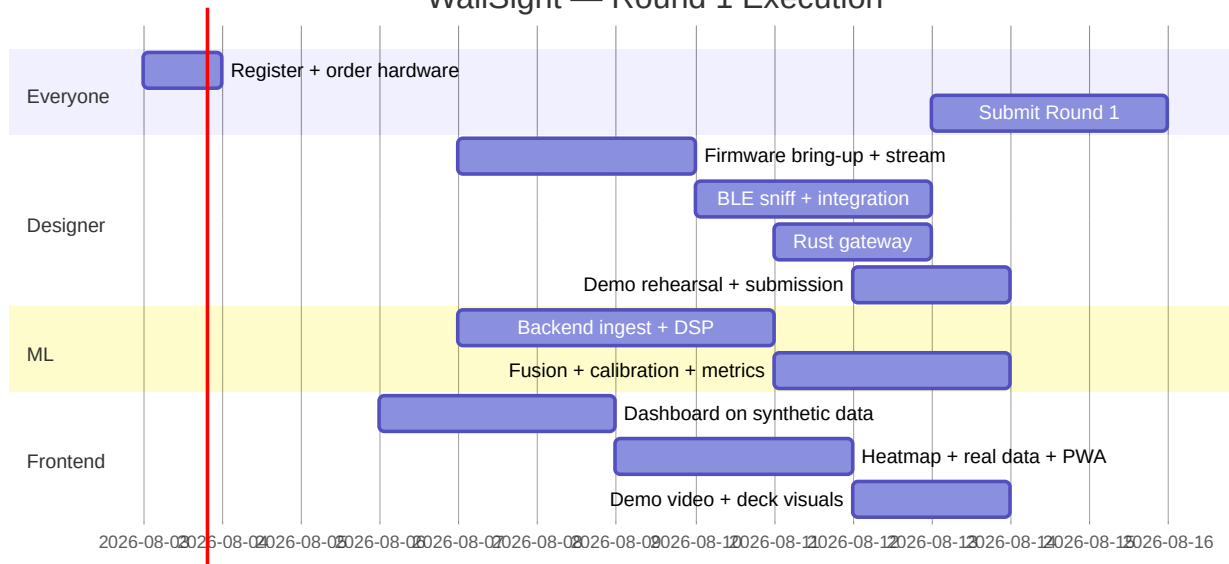
Coordination rules:

- Daily 15-min sync; everyone's deliverable is independently demoable until integration day
- Frontend builds against synthetic data from day 1 — never blocks on hardware
- One shared GitHub repo, one architecture doc (this file)

15. Timeline

Date	Milestone
Aug 3	✅ Order boards NOW (3–5 day shipping)
Aug 3–4	✅ Register at softtechitsolution.in/register (Unstop listing closes Aug 5)
Aug 6–8	Boards arrive · flash esp-csi firmware · CSI streams to laptop
Aug 9	Backend ingest API consumes live CSI · dashboard shows synthetic data
Aug 10	Motion detected from live stream · stream format frozen
Aug 11	Fusion engine (CSI+BLE) · breathing waveform live · accuracy numbers
Aug 12	Full system: 2 zones + heatmap + BLE + alerts · rehearsal · record demo
Aug 13–15	Round 1 submission (video + repo + deck + dashboard link)
Aug 21–22	Round 2 evaluation (if advanced)
Aug 28–30	Round 3 finals (if advanced)

WallSight — Round 1 Execution



16. Competition

What exists (we surveyed ~30 domains)

Domain	Existing	Verdict
Digital-arrest scam detection	CyberDudeBivash triage	Saturated — avoided
Emergency vehicle preemption	SmartEVP+	Saturated — avoided
Photo provenance (C2PA)	TrueShot	Saturated — avoided
Cold-chain tamper boxes	ColdProof, LogiChain360	Saturated — avoided
E-nose food freshness	FoodSense-AI	Saturated — avoided
AI stethoscope	RevoScope, CardioScan	Saturated — avoided
IMU sports analytics	multiple papers/projects	Saturated — avoided
EMG gesture	EMGBand, MyoPilot	Saturated — avoided
mmWave fall detection	CircuitDigest proj, Veriprajna	Saturated — avoided
Acoustic pipe leak	Sentinel, AquaSense, Leakio	Saturated — avoided
ESP32 smart-home firewall	T800, ShadowSentryS3	Saturated — avoided
UWB indoor positioning	tutorials + ₹2,500/modules	Too expensive — avoided
Elder/toddler safety as a consumer product (India) — tech proven by RuView, no one ships the product	Product lane open	We build this

Our direct lineage (algorithm-level)

- **RuView** (~83K stars, Rust-first): the platform that proves WiFi sensing works — presence, breathing, HR on ESP32 meshes, even a DensePose-like radio visualization. Free + open-source, edge-only, no cloud, no consumer app, no India presence → **we build the product it never built**
- **Origin Wireless**: commercial enterprise WiFi sensing — expensive, not consumer, not India

- **wifi-ghost / ThroughNet**: working open-source ESP32 builds we stand on

Why judges haven't seen this from students

Judges have likely seen RuView's tech demos — but shipping it as an **India-market consumer safety product** (caregiver app, cloud subscription, BLE fusion, ₹1,200 kit) is not something student teams present. We stand on RuView's shoulders honestly and differentiate on **product**, not raw tech.

17. Risks & Fallbacks

Risk	Probability	Mitigation
Boards arrive late	Med	Order Aug 3 (today); Delhi/UP electronics markets stock ESP32-S3; fallback = single board + home router mode
wifi-ghost firmware targets classic ESP32, not S3/C3	Med	Base firmware on <code>esp-csi</code> examples (chip-agnostic); port wifi-ghost's Python DSP only
CSI flaky in room	Med	Replay mode — demo runs recorded data; live is bonus
BLE + WiFi concurrency issues	Med	Time-shared scans; BLE is auxiliary not critical
Heart rate fails live	High	Breathing is the star; HR labeled experimental
ML guy's DSP port slips	Med	wifi-ghost ships runnable Python — port is configuration, not research
Team time conflict	Med	Independent deliverables until integration day; 15-min daily sync
Cloud tier limits	Med	Demo on localhost + screen recording; subscription architecture shown, not hosted at scale
Registration misses deadline	Low	Register before Aug 6 — done by designer day 1

18. Submission Checklist

- [] Register team "WallSight" at <https://softechitsolution.in/register> (**by Aug 4** — Unstop listing closes Aug 5)
- [] Boards ordered (**Aug 3**) and received (by Aug 8)
- [] Public GitHub repo: code + README + architecture diagrams (this doc becomes the README base)
- [] 5-minute demo video (scripted, replay-backed, edited)
- [] Pitch deck (slide outline below)
- [] Dashboard live link or screen recording
- [] Measured accuracy numbers from ML guy
- [] All teammates' contact details on the registration form

19. Pitch Deck Outline

1. **Hook** — one-liner + 10-second video of shadow-blob detection
2. **Problem** — the 2 AM wandering toddler / fallen grandma; participation-flaw of existing tech
3. **Physics in 30s** — "WiFi bounces; we read the bounces" (diagram)
4. **The 4 innovation pillars** — parasitic sensing · radio-shadow · ecosystem fusion · self-calibration
5. **Live demo** (video or recorded)
6. **Architecture** — dumb boxes → cloud brain → app (diagram)
7. **Accuracy + honest limits** — our numbers; "we know what we can't do"
8. **Business model** — subscription tiers, COGS, market size, ~90% margin
9. **Competition** — RuView/Origin exist at enterprise price; we're the consumer version; open lane
10. **Roadmap** — heart rate, OpenWrt nodes, triangulation, OTA
11. **Team** — designer · ML · frontend, why we're the right trio

20. Repos & Stack Reference

Stack

Layer	Tech
Firmware	ESP-IDF (C), espressif/esp-csi , adapted wifi-ghost firmware
Backend	Python FastAPI + NumPy/SciPy (DSP), ported wifi-ghost/ThroughNet algorithms
Real-time relay	Rust (WebSocket gateway) — designer's component
Frontend	React + TS, heatmap render (Canvas), zone editor
Mobile	PWA (installable, push alerts)
Cloud	Render/Railway free tier or localhost
Transport	MQTT/HTTP features streaming, WebSocket live updates

References

- Hackathon registration: <https://softechitsolution.in/register>
- Hackathon info: LT HackFest 2026 — LT Supercom India Pvt Ltd + EpochFolio (also on Unstop: "International Hackathon Competition 2026")
- esp-csi: <https://github.com/espressif/esp-csi>
- wifi-ghost: <https://github.com/heyfinal/wifi-ghost>
- ThroughNet: <https://github.com/Adichapati/ThroughNet>
- RuView: <https://github.com/ruvnet/RuView> (83K stars — presence/breathing/HR on ESP32, Rust core; study for reference)

Document status:  FINAL — approved by team lead. Update on integration-day findings.